

**10th International Conference on
Electrical Engineering, Computer Science and Information (EECSI 2023)**

Presentation on

**Discovering Novel Pharmaceutical Molecules with
Generative Adversarial Networks**

Aditi Kulkarni
Padmapriya Mohan

Namratha K
Raxna Bojamma P J

Dr. Leena Chandrashekar
Assistant Professor
Dept. of ECE, RNSIT

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Motivation

- Small molecule drug design is one of the first and the most important steps in drug discovery. However, due to novel molecule generation's complexity, drug design is both cost and time-inefficient.
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- Drug design is complex
 - Drug design is time inefficient
 - Drug design is expensive

Objectives

The objectives of this paper

- To develop a deep learning neural network for the generation of novel small molecules.
- To develop and train a GAN to discover new molecular configurations.
- To assess and validate the results with other benchmark techniques.

Literature Survey

Paper	Methodology	Challenges
Optimizing distributions over molecular space. An Objective-Reinforced Generative Adversarial Network for Inverse-design Chemistry (ORGANIC) [1]	Objective-Reinforced Generative Adversarial Networks (ORGAN) GAN and Reinforcement Learning (RL).	Generates a large amount of non-valid molecules. Failure lies when the density of molecules for a certain criteria is very sparse.
Using GANs with adaptive training data to search for new molecules [2]	Training GANs that promote incremental exploration and limit the impacts of mode collapse. Valid samples from the generator are used to replace samples from the training data.	One limitation is that a definition for valid generated samples must be given. The trade-off is that features of the original distribution may be lost as the training data shifts.

Paper	Methodology	Challenges
Relevant Applications of Generative Adversarial Networks in Drug Design and Discovery: Molecular De Novo Design, Dimensionality Reduction, and De Novo Peptide and Protein Design [3]	It depicts several studies in de novo peptide and protein design using GAN frameworks.	hard to converge and massively unstable when training the GAN architecture. the mode collapse problem in the GAN architecture.
Structure-based de novo drug design using 3D deep generative models [4]	DeepLigBuilder is a deep learning model. The Ligand Neural Network (L-Net) is a graph generative model. L-Net was combined with Monte Carlo tree search (MCTS) .	Non-closed aromatic rings will be marked as invalid molecules by RDKit. Standard deviation for the length of aromatic bonds from test set molecules is 0.051 Å but for generated molecules are 0.067 Å. The standard deviation of torsion angles is 1.72° for the generated molecules and 0.46° for the test set molecules.

Implementation

The following is the GAN algorithm used for our implementation:

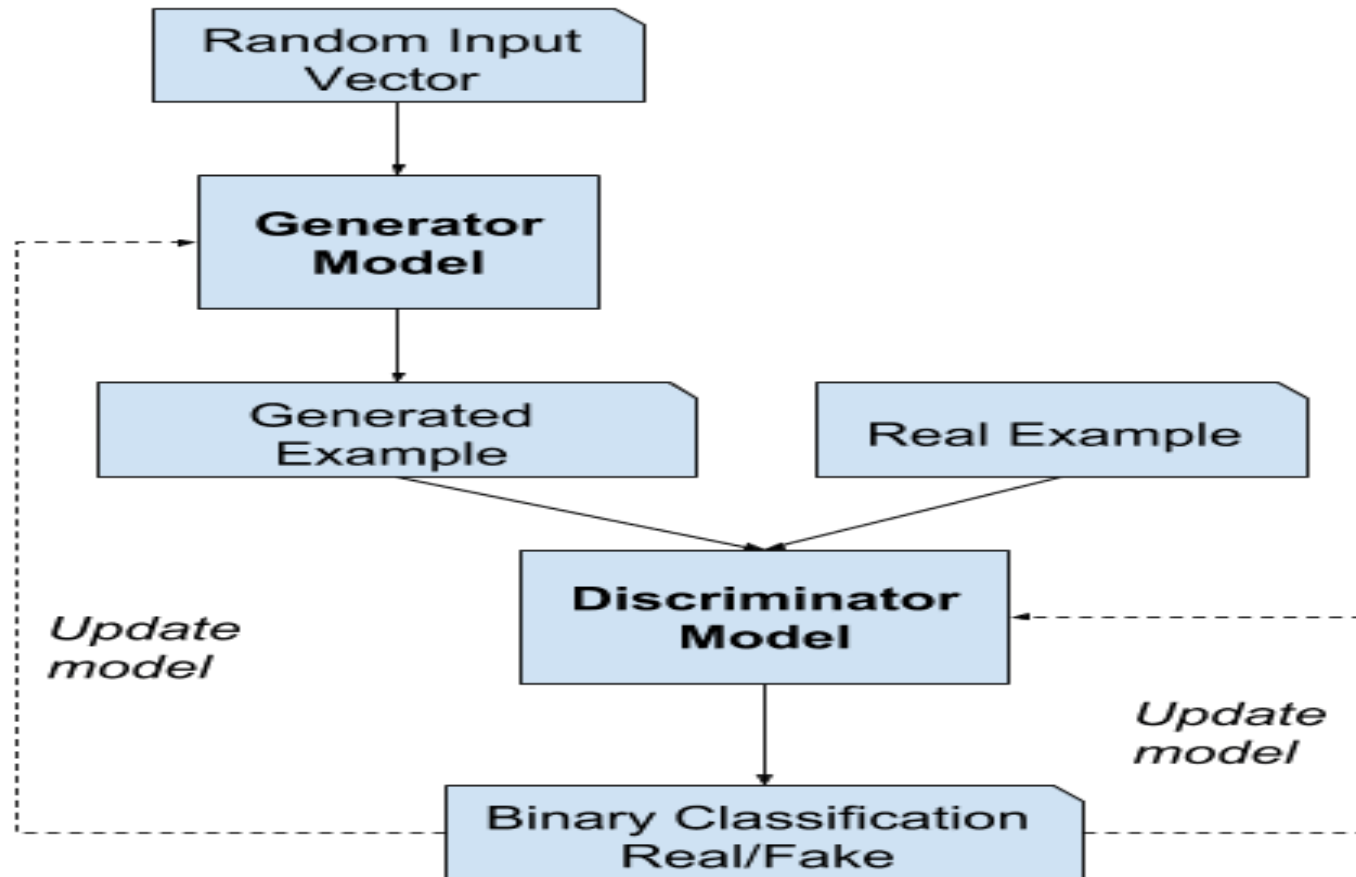


Image from <https://machinelearningmastery.com>

Methodology

The model consists primarily of a generator and a discriminator that compete against each other. The cost function is defined as below -

$$\min_{\theta} \max_{\phi} \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D_{\phi}(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\phi}(G_{\theta}(z)))] ,$$

Those two models are implemented as neural networks and trained simultaneously with stochastic gradient descent (SGD).

G_{θ} and D_{ϕ} have different objectives, and they can be seen as two players in a minimax game where G_{θ} tries to generate samples to fool the discriminator and D_{ϕ} tries to differentiate samples correctly.

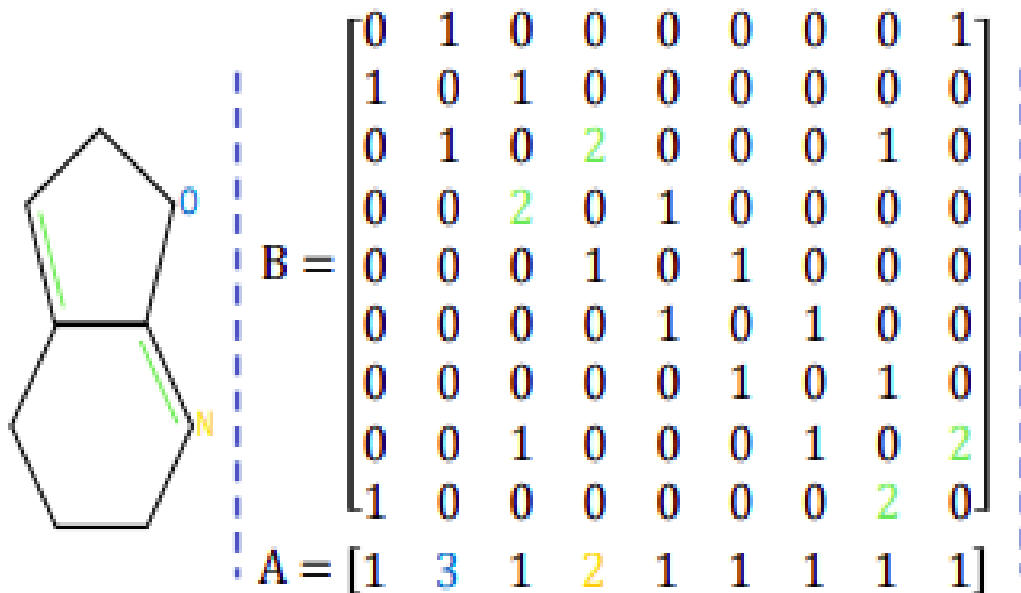
Dataset

- QM9 (Quantum Machines 9) provides quantum chemical properties for a relevant, consistent and comprehensive chemical space of small organic molecules
- This database serves as the benchmark of existing methods, development of new methods such as hybrid quantum machine learning and systematic identification of structure-property relationships.
- It consists of 134k smallest organic molecules containing up to 9 heavy atoms along with their quantum properties.
- The first line is the drug id, second line refers to the software package that created it. The next few lines describe the properties which state the number of atoms and bonds, each atom and bond on a separate line.

```
1 gdb_1
2 -OEChem-03231823243D
3
4 5 4 0 0 0 0 0 0 0999 V2000
5 -0.0127 1.0858 0.0080 C 0 0 0 0 0 0 0 0 0 0 0
6 0.0022 -0.0060 0.0020 H 0 0 0 0 0 0 0 0 0 0 0
7 1.0117 1.4638 0.0003 H 0 0 0 0 0 0 0 0 0 0 0
8 -0.5408 1.4475 -0.8766 H 0 0 0 0 0 0 0 0 0 0 0
9 -0.5238 1.4379 0.9064 H 0 0 0 0 0 0 0 0 0 0 0
0 1 2 1 0 0 0 0
1 1 3 1 0 0 0 0
2 1 4 1 0 0 0 0
3 1 5 1 0 0 0 0
4 M END
5 $$$$
6 gdb_2
7 -OEChem-03231823233D
8
9 4 3 0 0 0 0 0 0 0999 V2000
0 -0.0404 1.0241 0.0626 N 0 0 0 0 0 0 0 0 0 0 0
1 0.0173 0.0125 -0.0274 H 0 0 0 0 0 0 0 0 0 0 0
2 0.9158 1.3587 -0.0288 H 0 0 0 0 0 0 0 0 0 0 0
3 -0.5203 1.3435 -0.7755 H 0 0 0 0 0 0 0 0 0 0 0
4 1 2 1 0 0 0 0
5 1 3 1 0 0 0 0
6 1 4 1 0 0 0 0
7 M END
```

Methodology

The input to the generator is noise, whereas the input to the discriminator is a graph made of the atom vector and the bond matrix as shown below.



Atom types:

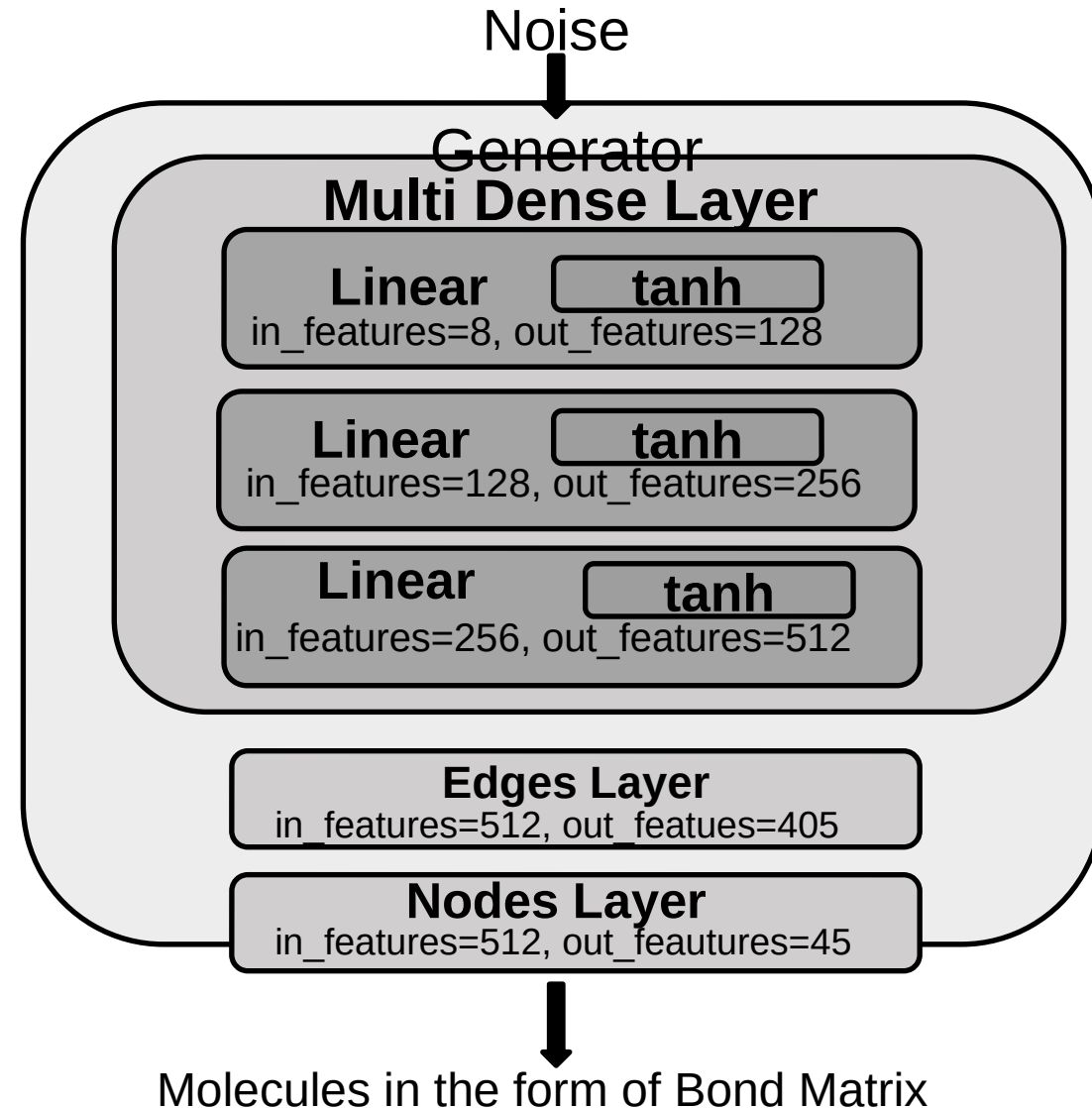
- 0- Padding symbol
- 1- Carbon
- 2- Nitrogen
- 3- Oxygen
- 4- Fluorine

Bond types:

- 0- No bond
- 1- Single bond
- 2- Double bond
- 3- Triple bond
- Aromatic

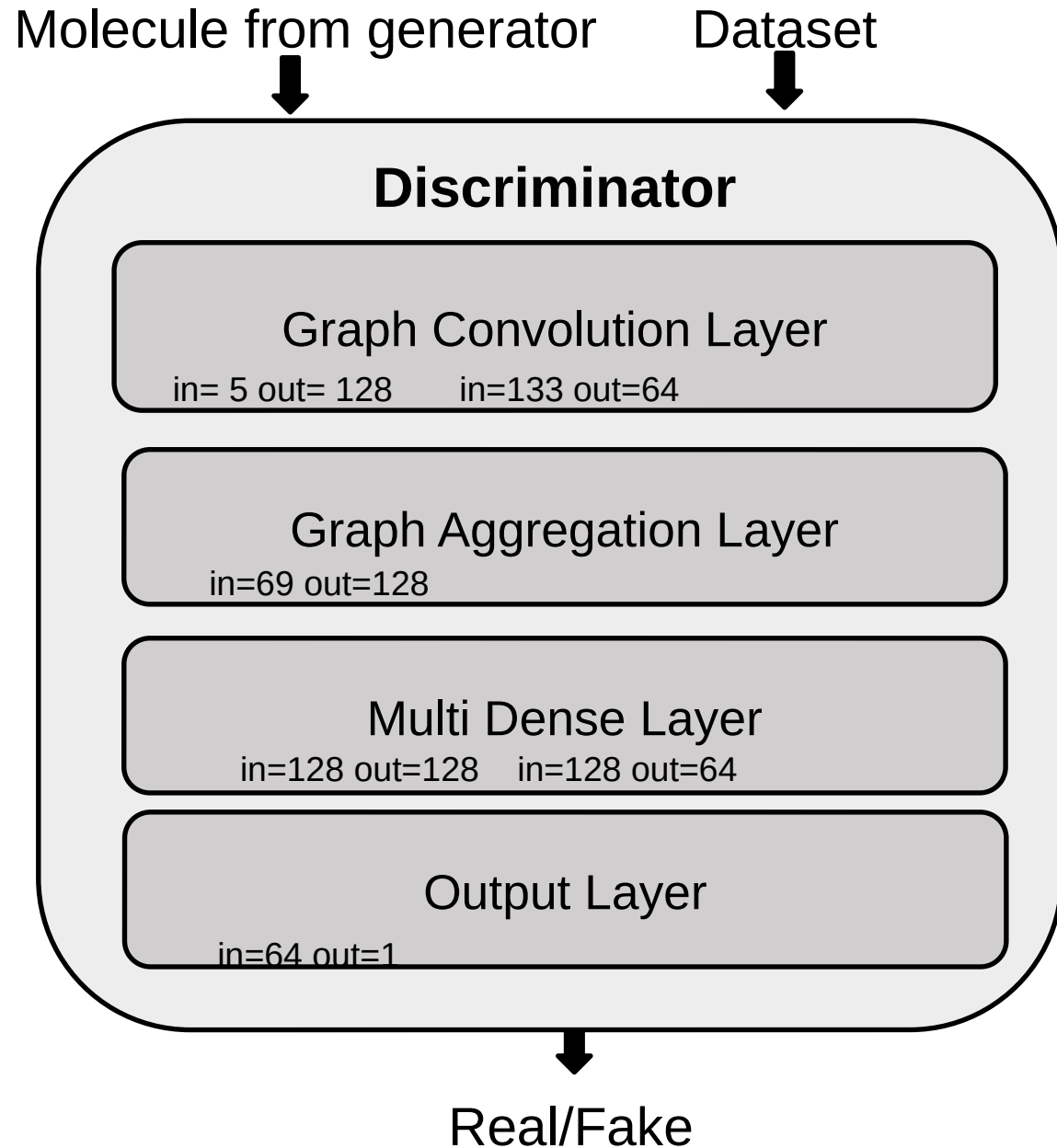
Architecture of Generator Network

- The generator consists of a 3-layer multi-layer perceptron (MLP) with 128, 256, and 512 hidden units.
- Tanh activation filter has a range of -1 to 1.
- The edges layer typically refers to the layer that processes the edges of a graph.
- The nodes layer applies a neural network function to each node in the graph, using the node features as inputs.



Architecture Of Discriminator Network

- GraphConvolution layers perform a convolution operation on the node features.
- GraphAggregation layers perform an aggregation operation on the neighboring nodes' hidden states.
- MLP consisting of two layers with 128 and 64 hidden units, respectively, further processes the representation and computes a final output score indicating the likelihood that the input graph is a valid molecule
- The output layer of the discriminator produces a single scalar value that represents the probability of the input molecule being real or fake.



Hyperparameter Optimization

In order to generate desired molecules and to optimally train our model, we use the following optimization strategy over the given list of hyperparameters -

List of hyperparameters considered:

- **Batch size-32**

Batch size is a hyperparameter which defines the number of samples taken to work through a particular machine learning model before updating its internal model parameters.

- **Learning rate - 0.0001**

It is a hyper-parameter used to govern the pace at which an algorithm updates or learns the values of a parameter estimate.

- **Number of epochs**

An epoch refers to the one entire passing of training data through the algorithm.

- **n_critic = 5 [7]**

It refers to the number of critic(discriminator) iteration per generator iteration.

- **Lambda_gp = 10**

The idea of Gradient Penalty is to enforce a constraint such that the gradients of the critic's output w.r.t the inputs to have unit norm.

Evaluation Metrics

- **Validity** - the ratio between the number of valid molecules and all the generated molecules. $\text{Validity} = |\text{Cs}|/n_s$
- **Novelty** - the ratio between the set of valid samples that are not in the dataset and the total number of valid samples.

$$\text{Novelty} = 1 - |\text{Cs} \cap D|/|\text{Cs}|$$

- **Uniqueness** - the ratio between the number of unique valid samples and the total number of generated samples. $\text{Uniqueness} = |\text{set}(\text{Cs})|/n_s$
- **QED(Quantitative estimate of drug-likeness)**- It is an integrative score to evaluate compounds' favourability to become a hit. It is a method to quantify the drug-likeness considering the main molecular properties together.

$$QED = \exp \left(\frac{1}{n} \sum_{i=1}^n \ln d_i \right)$$

Evaluation metrics

- **Synthetic accessibility(SA)** - a process to assess the ease of synthesis of compounds. It is dependent on the stability of the molecule's chemical composition which determines how the molecule exists physically.
- **Solubility - Solubility** is the degree to which a molecule is hydrophilic.
- **NP** - A natural product (NP) likeness score is a molecule's similarity to the structure space covered by NPs compared to the structure space covered by synthetic molecules (SM).
- **Diversity score** - It indicates how likely a molecule is to be diverse with respect to the dataset.

Results

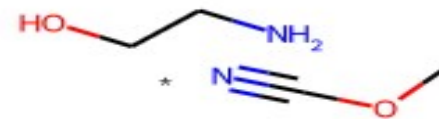
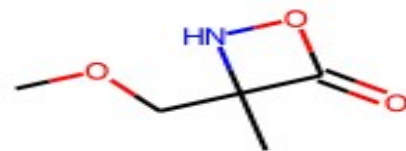
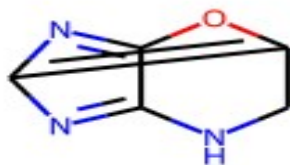
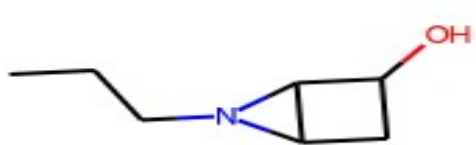
Number of epochs = 300 Dataset= QM9 with 5k molecules

Model	Validity	Uniqueness	Novelty	Solute
MolGAN [12]	87.7	2.9	97.7	0.54
Proposed GAN	95.6	5.5	98.4	0.384

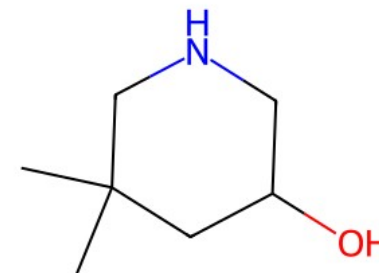
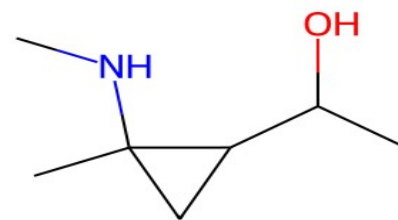
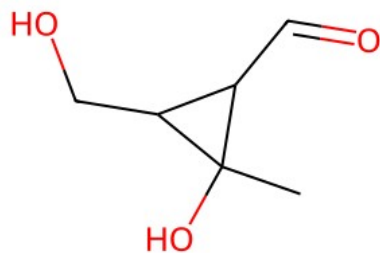
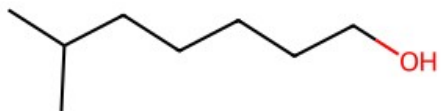
Model	Validity	Time	Druglikeness	Solute
ORGAN [13]	96.1	10.2	0.52	0.543
Proposed GAN	95.6	5.8	0.5290	0.384

Results

Example molecules generated by existing MolGAN (left) and QuMolGAN (right):



Results for this implementation of GAN:



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THANK YOU